

Introduction to Models

BIOL 3401 Mount Royal University

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Learning Objectives

Today, we will start learning about statistical models. Linear models are one of the simplest and most versatile classes of statistical model. We will add more complexity to our analyses in the weeks to come.

After completing this topic, you should feel comfortable:

1. Distinguishing between a test of difference and test of relationship
2. Test the assumptions of your linear models
3. Evaluate your model's fit, and consider how you could improve it
4. Fit some models!

Introduction to modelling in R

The most basic form of modelling is a linear model. Two types of linear model you have already heard of are:

- ANOVA is a *test of difference*. We want to know whether two groups—categories of a variable, $\mathbf{X}$ —come from two different populations, based on a variable, $\mathbf{Y}$, measured on both groups.
- Linear regression is a *test of relationship*. We want to know whether a variable, $\mathbf{Y}$, varies along levels of $\mathbf{X}$.

We already talked about how these tests are different forms of the same basic model. The main distinction is the range of possible values of $\mathbf{X}$. In *tests of difference*, $\mathbf{X}$ can only take a few *discrete* values or categories. In *tests of relationship*, $\mathbf{X}$ can theoretically take *any* value from a minimum to a maximum. In the first part of this tutorial, we will explore this idea.

Set up your R workspace for the day

Set your working directory

Make sure we're setting our working directory!

```
# Set your working directory, then remove 'eval = F'!  
setwd()
```

Install and load packages

We are going to need to load tidyverse. We are also going to install a new package called `car`, which stands for “Companion to Applied Regression”.

Using the code block below, install the `car` package, then load `car` and `tidyverse`.

Load data

Our dataset for this topic (`lizard_data.csv`) comes from a study on lizard running speeds in different habitats. Each row is a measurement from a new lizard. Lizards are ectotherms, which means their physiological performance depends on the external temperature. We will be testing a number of hypotheses relating body size, temperature, age, sex, habitat, and running speed. The columns are defined as follows:

- `sex`: Sex of the lizard (female or male)
- `habitat`: The habitat where the lizard was captured (open or forest)
- `age_years`: The age of the lizard
- `body_size_mm`: The snout-vent length of the lizard, measured in millimeters
- `body_temp_C`: The temperature of the lizard's body, in degrees Celsius, at the time of its running trial
- `speed_m_s`: The running speed of the lizard in metres per second

Load the dataset by finishing the `read_csv()` function.

```
lizard_data <- read_csv()
```

Part 1: Tests of difference

We're first going to use the lizard data to understand tests of difference. We're going to test the hypothesis that *sex- and age-specific reproductive strategies in this species drive differences in growth trajectories and eventual body size*. This hypothesis has two testable predictions:

- Older lizards are larger younger lizards
- Male lizards are larger than female lizards

Part 1a: Visualize the data

Before we test our hypothesis, we want to visualize how the distributions of data differ between groups. One of the easiest ways to do this is with boxplots. Let's start with visualizing the sex differences.

```
ggplot(lizard_data, aes(x = sex, y = body_size_mm)) + geom_boxplot() + labs(x = "Lizard sex",
  y = "Body size (mm)")
```

Size differences between male and female lizards We can see a clear difference. It looks like we might be on the right track.

Size differences between young and old lizards Next, let's look for size differences between young and old lizards. If you look at the `age_years` variable, you'll see it's continuous numeric data. We want to test for differences between groups, so we'll need to categorize the ages.

Mini Challenge 1

Ecologists have classified this lizard species into three age groups:

- younger than 2 years: Juvenile
 - from 2-4 years: Prime-age
 - older than 4 years: Senescent
1. Create a new factor data type column in the data called "age_class" to distinguish among the age groups
 2. Create a boxplot comparing the distributions of body size among the age groups
 3. Call the new dataset `lizard_age_classes`

Use the code block below to code your answers!

```
#-----
```

Part 1b: Test assumptions

Before we test statistically for group differences, we need to confirm our data follows the assumptions of linear models. Those assumptions again:

- Independent errors
- Linear relationship
- Homogeneity of variance
- Normally distributed errors

Let's ignore the first two assumptions for now and focus on *homogeneity of variance* and *normally distributed errors*. In tests of difference, these assumptions refer to the *response variable*, which is `body_size_mm`.

There are two important functions for testing these assumptions:

- `leveneTest`: A function from the `car` package that computes a Levene's test for homogeneity of variance across groups. A p-value < 0.05 indicates the variances are likely different between groups, a violation of linear model assumptions.
- `shapiro.test`: A function in base R that tests whether a vector of data follows a normal distribution. A p-value < 0.05 indicates the data likely do not follow a normal distribution.

Test assumptions for size differences between sexes Let's see how this works for a test of difference in body size between sexes.

Test for homogeneity of variance

The `leveneTest` function takes two important arguments. `y` is the response variable, which can either be a vector or a column from a data frame. `group` is the variable defining the groups. It can either be a vector or column, but must be the same length as `y`.

```
leveneTest(y = lizard_age_classes$body_size_mm, group = lizard_age_classes$sex)
```

The function spits out a warning about “coercing `lizard_age_classes$sex` to factor”. This is just because in our data, `sex` is not a factor data type, which is required by the function. R is telling us it has done that for us.

In any case, p ($\Pr(>F)$) is > 0.05 , so we're good!

Test for normality

We want to make sure that `Y` (`body_size_mm`) is normally distributed at *all* values of `X` (`sex`). The `shapiro.test` function only takes the argument `x`, which is the vector or column defining the response variable. We're going to look at each value of `X` (female and male) separately by indexing the columns.

```
# Test for normality in male body size
shapiro.test(lizard_age_classes[lizard_age_classes$sex == "Male", ]$body_size_mm)

# Test for normality in female body size
shapiro.test(lizard_age_classes[lizard_age_classes$sex == "Female", ]$body_size_mm)
```

It looks like the body size variable also meets the assumption of normally distributed errors.

Mini Challenge 2

We haven't tested the assumptions of normality and homogeneity of variance for body size with respect to the new age class variable we created in `lizard_age_classes`. Check these assumptions now.

Use the code block below to code your answers!

Do the data meet our assumptions?

```
#-----
```

Part 1c: Test for size differences between sexes

Now that we've tested our assumptions, we can fit a linear model. First, we'll test for differences between the sexes.

ANOVA Let's start with ANOVA, which is just a linear model, as we know. We are going to use the `aov()` function. The arguments in this function are `data` (the data frame) and `formula`, the formula for the linear model, which includes the dependent variable (body size) and independent variable (`sex`) separated by a tilde (`~`). The format is: `dependent variable ~ independent variable` (e.g., `body_size_mm ~ sex`).

```
# Fit the ANOVA
lizard_sex_anova <- aov(body_size_mm ~ sex, data = lizard_age_classes)
# Summarize the results
summary(lizard_sex_anova)
```

It looks like there is a significant difference between the sexes in terms of body size. But how do we know from this output which sex is larger?

Linear model This is where the linear model comes in! Linear models are essentially equivalent to ANOVA in testing for significant differences between groups, but they have the additional advantage of providing an effect size that tells us the differences between groups. With the linear output model, we can see *exactly how much larger one sex is compared to the other*.

The function we will use for linear models is `lm()`. This function takes the exact same arguments as `aov()`—a formula and data frame.

```
# Fit the linear model
lizard_sex_lm <- lm(body_size_mm ~ sex, data = lizard_age_classes)
# Summarize the results
summary(lizard_sex_lm)
```

This output looks a little different than our ANOVA. Some additional things we can see:

- **Estimate:** This is the “*effect size*” of the variable. The effect size **sexMale** tells us how much larger males are than females. When the effects are categories, the model automatically picks a “reference category” for comparison, which is “Female” in this case. The estimate is, therefore, a prediction of how much larger body size is when the lizard is male versus female. The effect size for **(Intercept)** is the mean body size of the reference category, i.e., when the lizard is female.
- **Std. Error:** This is the variation or uncertainty in the effect size.
- **Pr(>|t|):** This is the p-value, telling us whether the relationship is statistically significant (< 0.05).

At the bottom of the summary, we can see values for **R-squared**. This is a measure of model “fit”, with larger values indicating better fit of the model to the data. There is an adjusted R-squared and multiple R-squared. We’ll talk more about the differences later. For now, these values are essentially equivalent.

Part 2: Tests of relationship

We tried to use test of difference to test the prediction that older lizards are larger than younger lizards, but the data did not meet our assumptions when we grouped the ages into categories. A *linear regression* is a more natural way to think about this test because age is essentially continuous data. It turns out that linear regressions are just like tests of difference, where the **X** variable can take *any* value, instead of just a few discrete values (e.g., Juvenile, Prime-age, Senescent).

Let’s revisit the prediction that lizard size increases with age.

Part 2a: Fit the linear regression

We are going to start backwards and first fit the linear regression, then test the assumptions. This is because R needs the actual linear model object to evaluate its assumptions.

Because every linear model is just a linear model, we’re going to use the same `lm()` function again. Again, the format is `dependent variable ~ independent variable`.

Mini Challenge 3

Let’s fit a separate model for male and female lizards to test the prediction that size increases with age.

1. Filter the data to create two lizard datasets, one with only males, and the other with only females. Call the datasets “M_lizard_data” and “F_lizard_data”.
2. Use the `lm()` function to test for an effect of `age_years` on `body_size_mm` for each sex.
3. Call the models “size_age_M_lm” and “size_age_F_lm”.

Use the code block below for your answer!

```
#-----
```

Let’s look at the summaries of the models.

```
# Male lizard model
summary(size_age_M_lm)
# Female lizard model
summary(size_age_F_lm)
```

Our output looks very familiar, but there are a few key differences. First, we now have an Estimate for `age_years`. Instead of comparing males and females, like our model of body size differences between the sexes, this estimate reflects *the amount by which body mass changes for each unit increase in age*. Because the estimate is positive, there is an increase in size with age, which we expect.

The meaning of (Intercept) has also changed. In our model comparing males and females, the intercept reflected the mean size of lizards in the reference category. *We now need to interpret the intercept as the lizard’s size when age = zero.*

We also see our R-squared, which tells us about the model’s fit.

Part 2b: Test assumptions

Now we will test whether the linear models we just fit meet the assumptions of linear models. This time, we are going to test *all* of the assumptions, including that of independent errors and linear relationship, which didn’t make as much sense in our test of difference. Again, the full suite of assumptions of linear regression are:

- Independent errors
- Linear relationship
- Homogeneity of variance
- Normally distributed errors

Instead of `shapiro.test()` and `leveneTest()`, we are going to evaluate our assumptions visually. The `plot()` function, when using a linear model object as the argument, provides a series of diagnostic plots that help us assess the assumptions of linear models. When you run the function, the plots will appear in the “Plots” window. In your console, R will tell you to “Hit to see next plot:”. The first two plots are:

1. **Residual plot:** Shows us whether the relationship is linear, whether there are any patterns in the errors that could cause non-independence, and whether the variances are homogeneous.
2. **Quantile-quantile (Q-Q) plot:** Shows us whether the errors are normally distributed.

Let’s plot the female model first:

```
plot(size_age_F_lm)
```

This looks decent. In the residual plot, the fact that the points are spread out around zero shows us three of our four assumptions are met. If the model violated these assumptions, we would see a pattern in the points.

The Q-Q plot shows us that the errors are normally distributed. If they weren't, we might see either an s-curve or a parabola shape around the diagonal line.

Let's test the male model:

```
plot(size_age_M_lm)
```

This also looks good!

Test the relationship between running speed and temperature.

Next, we are going to test the hypothesis that external temperature affects physiological performance in this species. As an ectotherm, cooler temperatures should reduce performance in our lizard. Our associated prediction is that lower body temperatures will result in slower running speeds.

Mini Challenge 4

1. Fit a linear model to test the relationship between `speed_m_s` and `body_temp_C`, *including both male and female lizards in the same model*.
2. Use your diagnostic plots to assess whether the model meets the assumptions of a linear model.

Use the code block below for your answer!

Does your model meet the assumptions? If not, which assumptions do you think have been violated?

#_____

Part 3: Testing for outliers

Outliers are data points that fall far outside of the expected pattern of the regression model, potentially impacting results and model interpretations. Because they are influential, we need to be able to assess their impact and remove them if necessary.

We are going to test statistically for outliers. However, best practice is to *remove outliers only if they have undue influence over the model results and we have reason to believe they don't reflect the true variation in the data*. Sometimes we see some extreme data points, but that doesn't necessarily mean there is anything wrong. Still, we want to be able to compare the model results with and without the outliers to evaluate their impact on our conclusions.

But how do we know which data points are potential outliers?

We will use the function `cooks.distance()` to calculate the Cook's distance index on all individual data points in our regression. This statistically tests the influence of individual data points on our results. There is no strict definition for what we mean by "influential", but as a rule of thumb, a data point could be significantly influential if its computed Cook's Distance is $\geq 4/\text{number of rows in the data}$.

Part 3b: Calculate Cook's distance

Let's check for outliers in our linear model we made to test the relationship between size and age in *male* lizards (`size_age_M_lm`). First, we need to calculate Cook's distance. We are going to save the output as a tibble so we can work with the values.

```
cooks_D_size_age_M <- as_tibble(cooks.distance(size_age_M_lm))
```

Next, we'll calculate the cut-off for Cook's distance values being potentially influential. We can use the function `nrow()` to determine how many rows are in our tibble with male lizard data, `M_lizard_data`.

```
4/nrow(M_lizard_data)
```

So, any data points with Cook's distance values greater than approximately 0.021 are potentially influential.

The row numbers in our new tibble `cooks_D_size_age_M` correspond to the same rows in `M_lizard_data`. This means we can isolate the specific rows—i.e, data points—in `cooks_D_size_age_M`, with Cook's distance values ≥ 0.021

Mini Challenge 5

1. Using vector indexing, find and remove the potentially influential data points from the `M_lizard_data` tibble.
2. Fit the linear model again to test the relationship between size and age in male lizards.

Use the code block below for your answer!

Have the model results changed? Would you recommend removing the outliers or leaving them in your model?

```
#-----
```

Part 4: Multiple regression

So far, we've only tested the relationship between two variables in one model. However, linear models are not just limited to a single relationship. We can add *additional independent variables* to our linear models. More variables generally explain more variation in the independent variable.

Based on the R-squared values from our model summaries, we know that both sex and age individually explain some of the variation in body size. How much more variation in body size would we explain with both variables in the same model?

To add multiple variables to a model, we use the `+` operator on the right side of the `~` in our formula. E.g., `response ~ variable1 + variable2 + ... variable_n`. We can technically add as many variables as we want. However, we also need to be careful. Adding too many variables can lead to something called **overfitting**, which actually makes it *more difficult to understand the patterns in our data*.

We will talk more about overfitting in a couple of weeks. For now, you just need to think about being as parsimonious as you can. Which variables do you need to include to address the hypotheses you are trying to test? Including age and sex in the same model makes sense for us, as our hypothesis is that *sex- and age-specific reproductive strategies in this species drive differences in growth trajectories and eventual body size*. Let's compare how our model changes when we add sex and age to the same model.

Effects of sex and age on body size

Here are our original three models.

```
# Effect of sex on body size
summary(lizard_sex_lm)
# Effect of age on male body size
summary(size_age_M_lm)
# Effect of age on female body size
summary(size_age_F_lm)
```

Individually, each of these models explains between approximately 33% and 67% of the variation in body size (the definition of R-squared). What if we included sex and age in the same model?

```
# Multiple regression model including sex and age effects on body size
lizard_sex_age_lm <- lm(body_size_mm ~ sex + age_years, data = lizard_data)
# Model summary
summary(lizard_sex_age_lm)
```

Our R-squared is now 0.78, which means this model explains a full 78% of the variation in body size. That's a great model!

Interactions between sex and age

Our model includes two independent variables, each having an individual effect on body size. However, it could be that the two variables have some *interactive* effect. What if we believe, for instance, that females grow faster than males as they age? Our model doesn't capture that possibility because the two independent variables are included separately. However, we can specify an *interaction* in our model. This will effectively estimate:

- How much bigger males are than females
- How quickly the lizards grow for each unit change in age
- How much faster the lizards grow, for each unit change in age, when they are females

To help you visualize what we're talking about, let's make a quick scatterplot of the change in body size with age for male and female lizards.

```
ggplot(lizard_data, aes(x = age_years, y = body_size_mm, colour = sex)) + geom_point() +
  geom_smooth(method = "lm") + labs(x = "Lizard age (years)", y = "Body size (mm)")
```

We can see there is a positive relationship between age and body size for both sexes. However, female lizards appear to grow faster than males, even though they start at a smaller body size and eventually reach a smaller adult size.

Let's see if accounting for the interactive effect explains any more variation in body size. We can include an interaction using the `*` operator. Using this operator still specifies the additive effects of each variable, but includes the interaction in addition. This interaction can be thought of as the amount by which the additive effect changes when the lizard is a male instead of a female.

Mini Challenge 6

1. Fit a new linear model including an interaction between `sex` and `age_years`
2. View the model summary

Use the code block below for your answer!

What has changed in our model summary? Does there appear to be an important interaction between age and sex?

#_____

Part 5: Model selection

The R-squared for this model with the interaction is also slightly higher, meaning the interaction explains slightly more variation in body size! But how do we know that the model with the interaction is worth it? We already talked about the principle of parsimony. We also saw that adding the interaction technically adds a new variable, which increases the risk that our model is overfit. How can we weigh the risk of overfitting the model against the benefit of the interaction?

We have now entered the realm of model selection procedures. Model selection allows us to weigh the relative support for models when we have multiple options. The procedure we are going to use is called the **Akaike Information Criterion (AIC)**. We aren't going to go into too much detail about the theory of AIC, but you can think of it as a measure of how the model compares to an unknown *true* model that perfectly describes 100% of the variation in the response we are estimating (`body_size_mm`). Such a model would theoretically have an R-squared = 1 and the linear regression would be a perfect slope = 1. This is obviously unrealistic. However, *the lower the AIC, the closer our model is to this idealized model*. AIC also *penalizes models for each additional parameter*, allowing us to evaluate how much better the model gets while accounting for more parameters. Generally, if one model is ≥ 2 AIC lower than the model with the next-lowest AIC, it is considered a superior model.

Mini Challenge 7

The function to calculate AIC for a model is just `AIC()`. The only required argument is the model object from your environment.

1. Using `lm()`, fit the following four models. *Don't worry about separating male and female lizards—include both sexes in the same models.*
 - `body size ~ age`
 - `body size ~ sex`
 - `body size ~ age + sex`
 - `body size ~ age * sex`
2. Evaluate each model using AIC

Use the code block below for your answer!

Which model is the “best”, model according to AIC?

#_____